

Switching Circuits and Logic Design (CS21202)

Module 9

Digital/Analog and Analog/Digital Converters

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1

Digital-to-Analog Converter

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2

2

Interfacing with the Analog World

- The physical parameters that exist in the external world are mostly analog in nature.
- The logic design principles we discussed so far focus on digital signals and digital circuits.
- To design a complete system that can interface with the external world, we must:
 - a) Convert analog signals to digital signals
 - b) Convert digital signals to analog signals

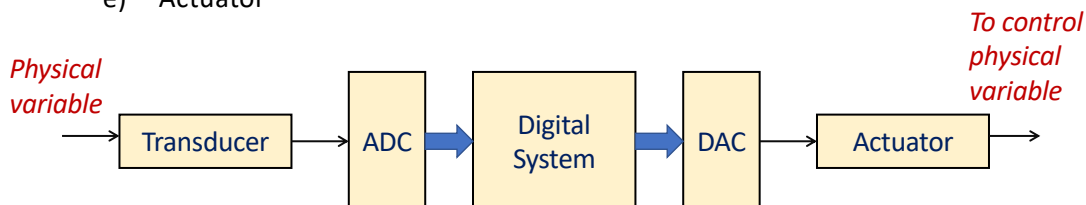
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Interfacing with the Analog World (contd.)

Required building blocks:

- a) Transducer: converts physical variable to electrical variable.
- b) Analog-to-digital converter (ADC)
- c) Digital System (digital circuit or computer)
- d) Digital-to-analog converter (DAC)
- e) Actuator



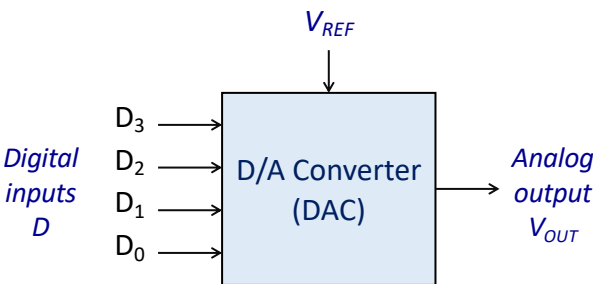
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Digital-to-Analog Converter (DAC)

- Converts a given digital word D to a proportional analog voltage V_{OUT} .

$$V_{OUT} \propto D$$

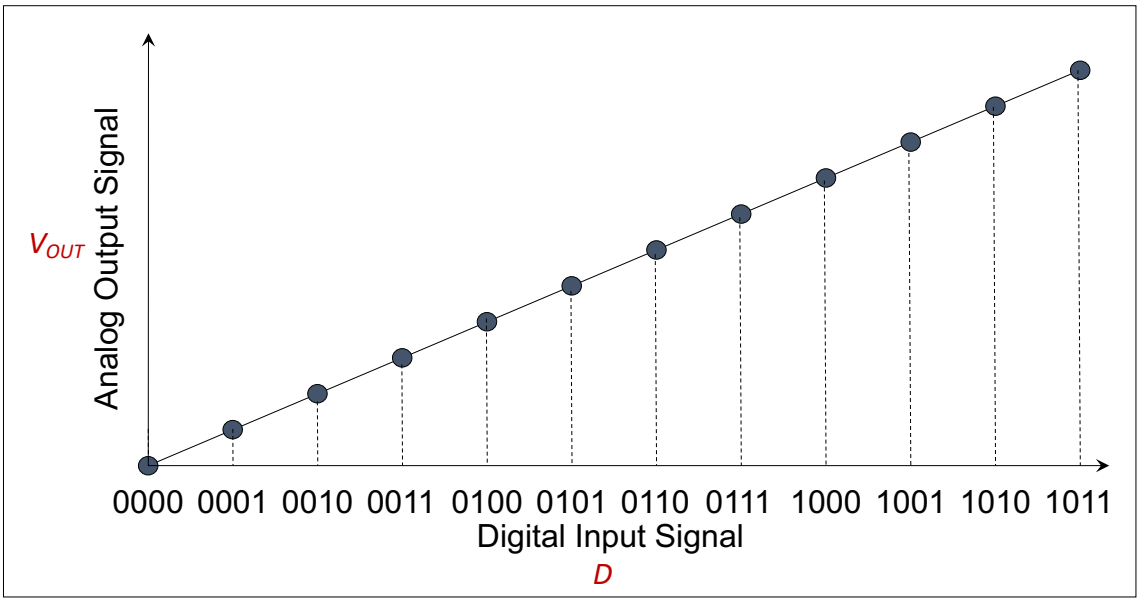


$$V_{REF} = 15 \text{ V}$$

D ₃	D ₂	D ₁	D ₀	V _{OUT}
0	0	0	0	0 V
0	0	0	1	1 V
0	0	1	0	2 V
0	0	1	1	3 V
0	1	0	0	4 V
0	1	0	1	5 V
0	1	1	0	6 V
0	1	1	1	7 V
1	0	0	0	8 V
1	0	0	1	9 V
1	0	1	0	10 V
1	0	1	1	11 V
1	1	0	0	12 V
1	1	0	1	13 V
1	1	1	0	14 V
1	1	1	1	15 V

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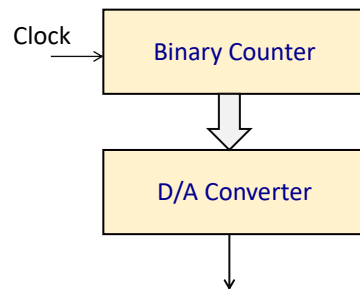


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Generating Staircase and Triangular Waveforms

Using a counter and a D/A converter.

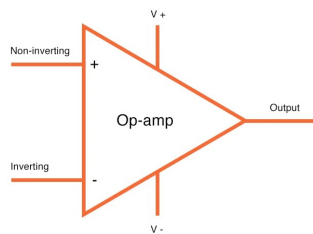


7

7

Operational Amplifier

- An operational amplifier, or op-amp, is a high-gain, DC-coupled circuit component that amplifies the voltage difference between two input terminals (inverting and non-inverting) to produce an output signal.



$$V_{OUT} = \beta (V_{IN}^{+} - V_{IN}^{-})$$

Ideal characteristics:

- Infinite open-loop gain
- Infinite input impedance
- Zero output impedance

Practical values are close to the ideal

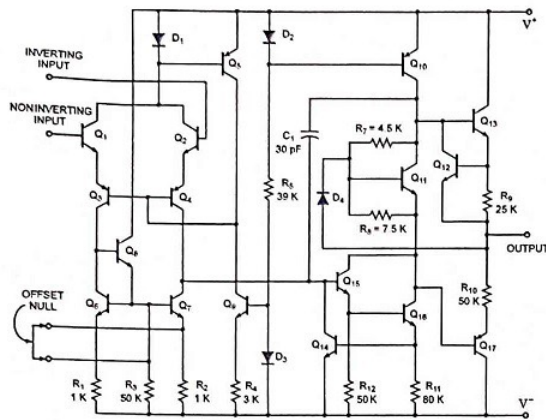
Typical applications:

- Variable-gain amplifier
- Voltage comparator
- Voltage buffering
- Mathematical operations

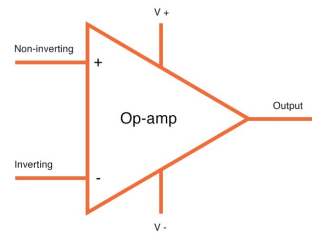
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8

8



Internal Circuit of LM741 opamp



- Since the open-loop gain is very high and the output voltage can only be finite (limited by V^+ and V^-), the voltages at the '+' and '-' terminals are almost equal.
- For instance, if the '-' terminal is grounded, the '+' terminal will also be at ground potential (called virtual ground).

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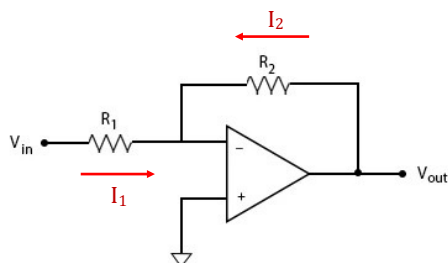
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9

Opamp Amplifier Circuits

• Inverting Amplifier:

- An inverting opamp amplifier is a circuit with an output voltage that changes in the opposite direction to the input voltage. In other words, it is out of phase by 180° .



Since the '-' terminal is at virtual ground,

$$I_1 = V_{in} / R_1$$

$$I_2 = V_{out} / R_2$$

As $I_2 = -I_1$, we can write

$$V_{out} / R_2 = -V_{in} / R_1$$

$$V_{out} = -(R_2 / R_1) V_{in}$$

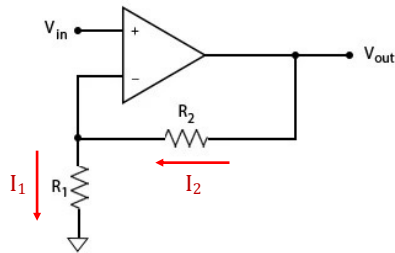
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10

10

• Non-inverting Amplifier:

- A non-inverting op-amp amplifier is a circuit whose output voltage is in phase with the input voltage.



Since the '-' terminal is at voltage V_{in} ,

$$I_1 = V_{in} / R_1$$

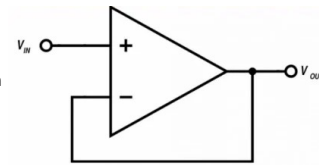
$$I_2 = (V_{out} - V_{in}) / R_2$$

As the input impedance is very high, $I_2 = I_1$, so

$$V_{in} / R_1 = (V_{out} - V_{in}) / R_2$$

$$V_{out} = (1 + R_2/R_1) V_{in}$$

If $R_2 = 0$, and R_1 is infinity (open circuit), then $V_{out} = V_{in}$
The circuit acts as a voltage buffer.

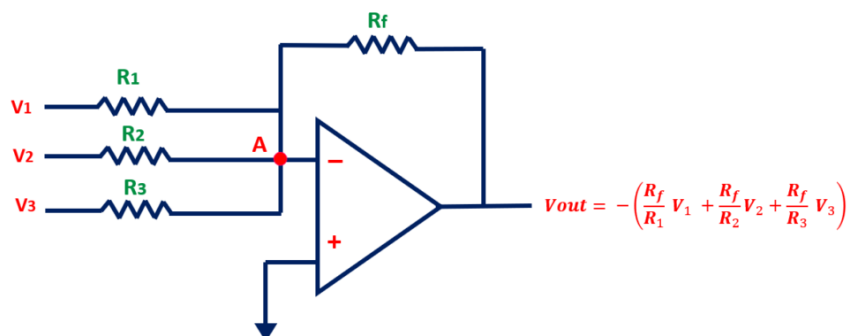


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11

11

• Weighted sum of voltages using op-amp:

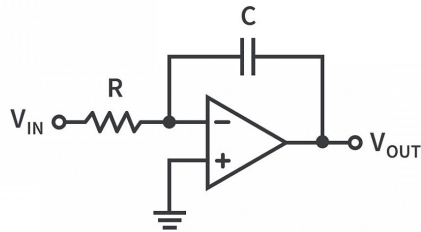


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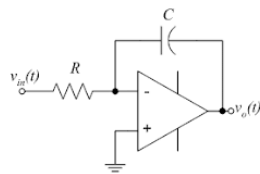
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- **Opamp Integrator:**

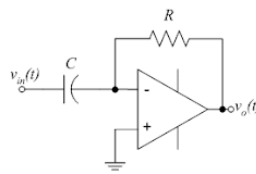


$$V_{OUT} = -\frac{1}{RC} \int_0^t V_{IN}(t) dt$$



(a) ideal integrator

$$v_o(t) = -\frac{1}{RC} \int_0^t v_{in}(t) dt + v_o(0)$$



(b) ideal differentiator

$$v_o(t) = -RC \frac{dv_{in}(t)}{dt}$$

Interchange R and C; you get a voltage differentiator

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13

13

Use of op-amp in Analog Computers

- **Solving a differential equation**

- Using op-amp circuits, we can implement:

- A voltage amplifier
- A voltage adder
- A voltage inverter (changes the polarity) – inverting mode with $R_1 = R_2$
- An integrator
- A differentiator
- A scalar multiplier (multiply a voltage source with a scalar)
- A circuit to compute the weighted sum of several voltage sources

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14

14

Basic steps to be followed

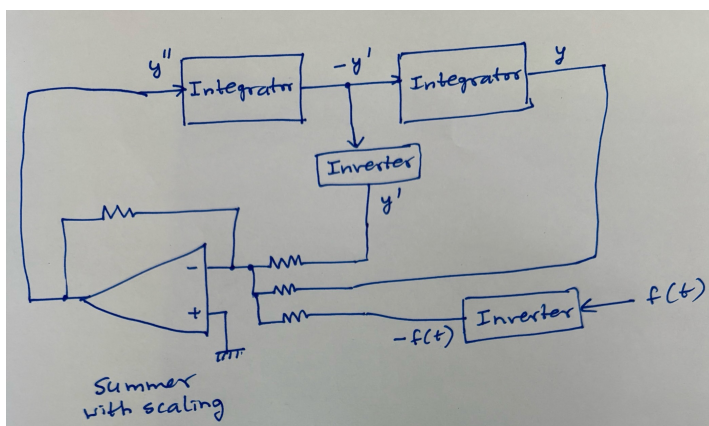
- Isolate the highest-order derivative (y'') on one side, and move all other terms to the other side.
- Assuming the highest derivative (y'') is available, feed it into integrators to obtain lower-order derivatives y' and y .
- Use inverting op-amp integrators to get the negative of the derivative (e.g., inputting $-y''$ to an integrator produces y').
- Use a summer to combine the signals and generate the highest-order derivative based on the rearranged equation.
- Select R and C values to match the coefficients of the differential equation.

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15

- Example: $a \frac{d^2y}{dt^2} + b \frac{dy}{dt} + cy = f(t)$
- Rearrange as: $\frac{d^2y}{dt^2} = -\frac{b}{a} \frac{dy}{dt} - \frac{c}{a} y + \frac{1}{a} f(t)$



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16

16

Some Parameters of DAC

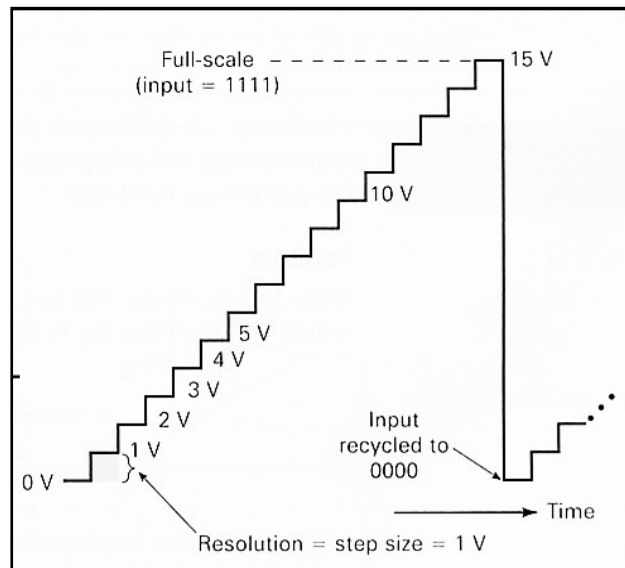
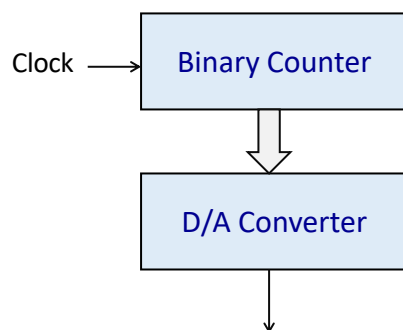
a) Resolution or step size:

- Smallest change that can occur in V_{OUT} as a result of a change in input D .
 - Equal to the weight of the LSB, also called *step size*.
 - Same as the constant of proportionality in $V_{OUT} \propto D$
- Can also be defined as a percentage of the full-scale voltage:

$$\begin{aligned}\% \text{ resolution} &= \text{step size} / V_{REF} \times 100 \\ &= 1 / (\text{total number of steps}) \times 100 \\ &= 1 / (2^N - 1) \times 100, \text{ for an N-bit DAC}\end{aligned}$$

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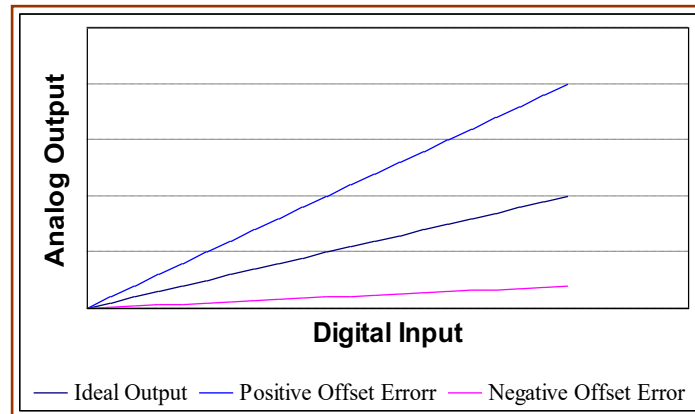


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b) Gain Error:

- Occurs when the slope of the actual output deviates from the slope of the ideal output.

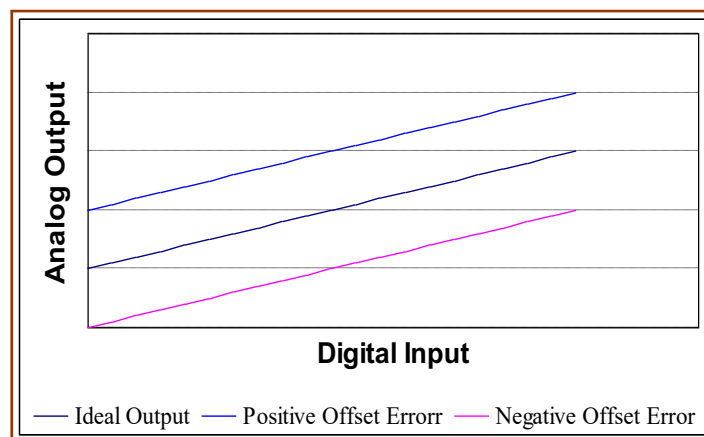


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c) Offset Error:

- Occurs when there is a constant offset between the actual output and the ideal output.



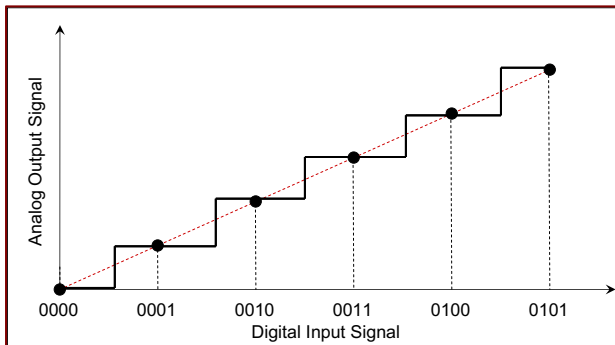
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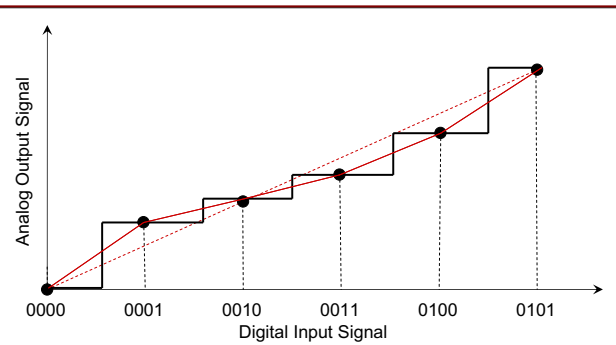
d) Non-Linearity Error:

- Occurs when analog output of signal is non-linear.

Linearity (Ideal)



Non-Linearity

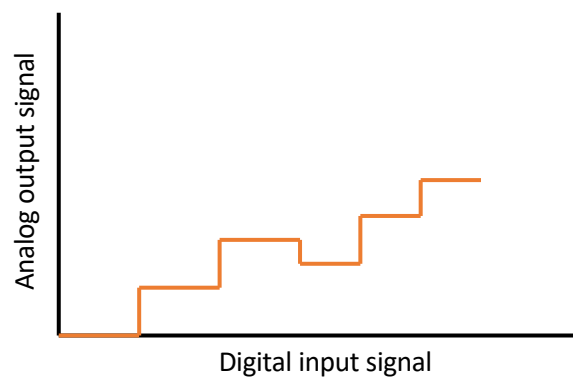


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21

e) Monotonicity Error:

- Occurs when an increase in digital input results in a decrease in analog output.



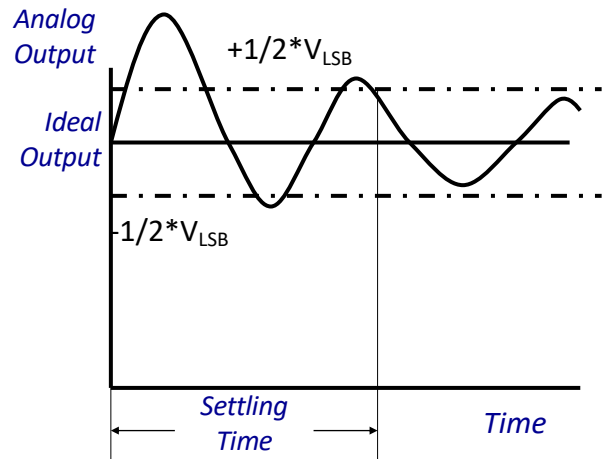
Suppose one of the inputs to the DAC is permanently connected to 0 due to some fault.

22

22

f) Settling Time and Overshoot Error:

- Settling Time is the time required for the output to fall within $\pm \frac{1}{2} V_{LSB}$.
- Overshoot occurs when analog output overshoots the ideal output.



23

23

Types of DACs

- We shall discuss two different designs of digital-to-analog converters.
 - Weighted resistor type DAC
 - Resistive ladder type DAC
- The first type is easier to analyze, while the second type is more practical from the point of view of implementation.

24

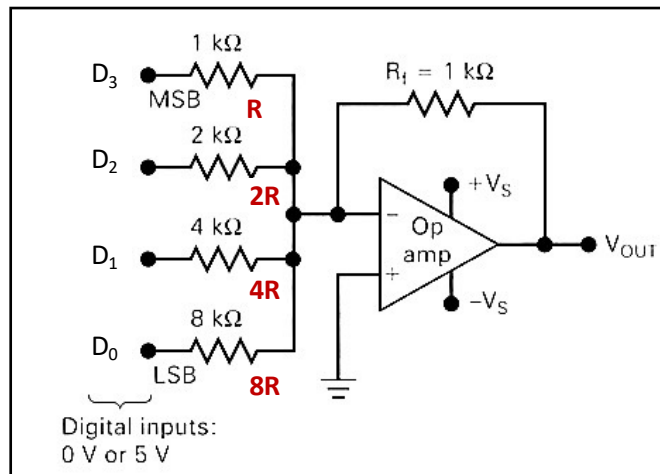
24

Weighted Resistor Type DAC

- An n -bit DAC consists of n different resistance values of magnitudes $R, 2R, 4R, \dots, 2^{n-1}R$, respectively.
 - The resistances help in generating currents inversely proportional to their magnitudes.
 - The total current is summed by an *operational amplifier* and converted into the output voltage V_{OUT} .
- Main drawback:
 - A precision resistor with a different value must be used for each bit position of the digital input.

25

25



4-bit weighted-resistor D/A converter

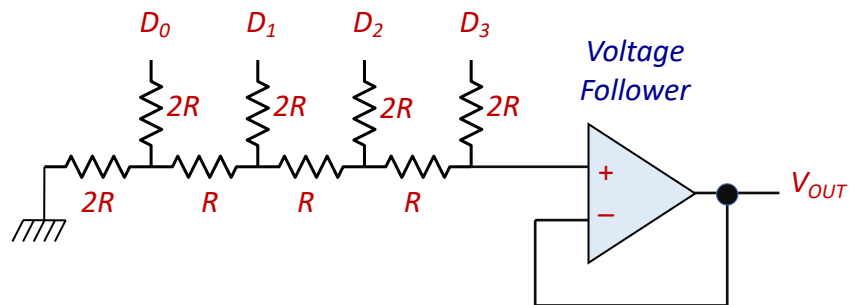
$$\begin{aligned}
 V_{OUT} &= - [D_3 + D_2 / 2 + D_1 / 4 + D_0 / 8] R_f / R \\
 &= - [8D_3 + 4D_2 + 2D_1 + D_0] R_f / 8R \\
 &\propto D
 \end{aligned}$$

26

26

Resistive Ladder Type DAC

- Most widely used, and requires only two different values of precision resistances (R and $2R$).

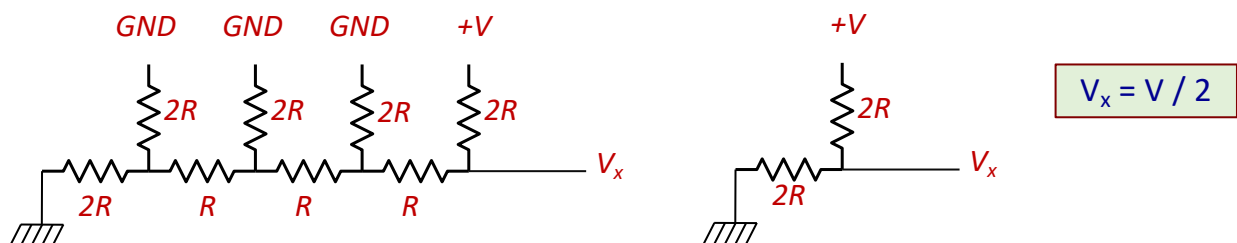


27

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Calculation

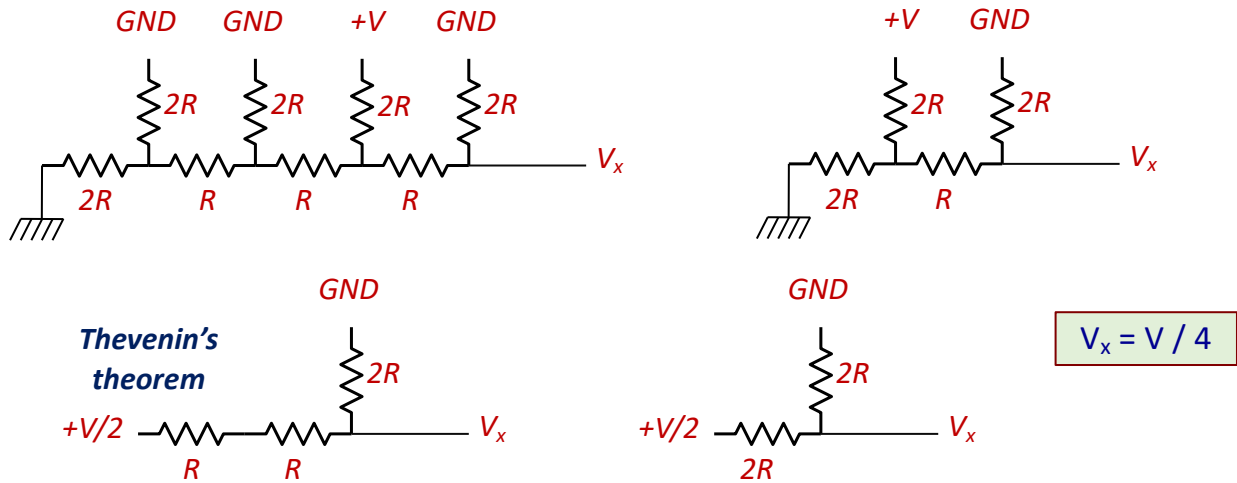
- Let us calculate the voltages at the opamp input when exactly one of the D_i inputs is at 1 (say, $+V$ volts).
- Case 1: Input is 1000.



28

28

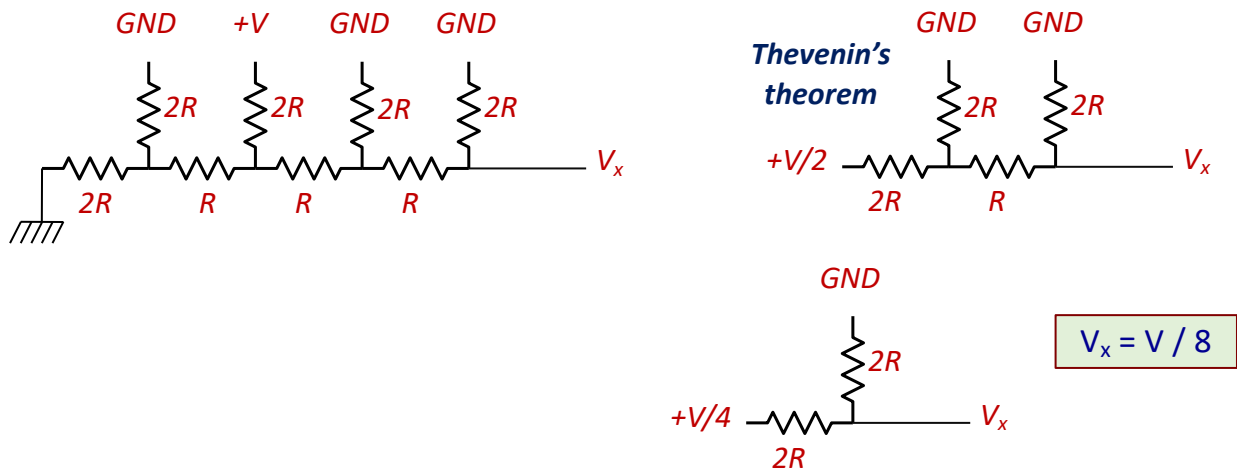
- Case 2: Input is 0100.



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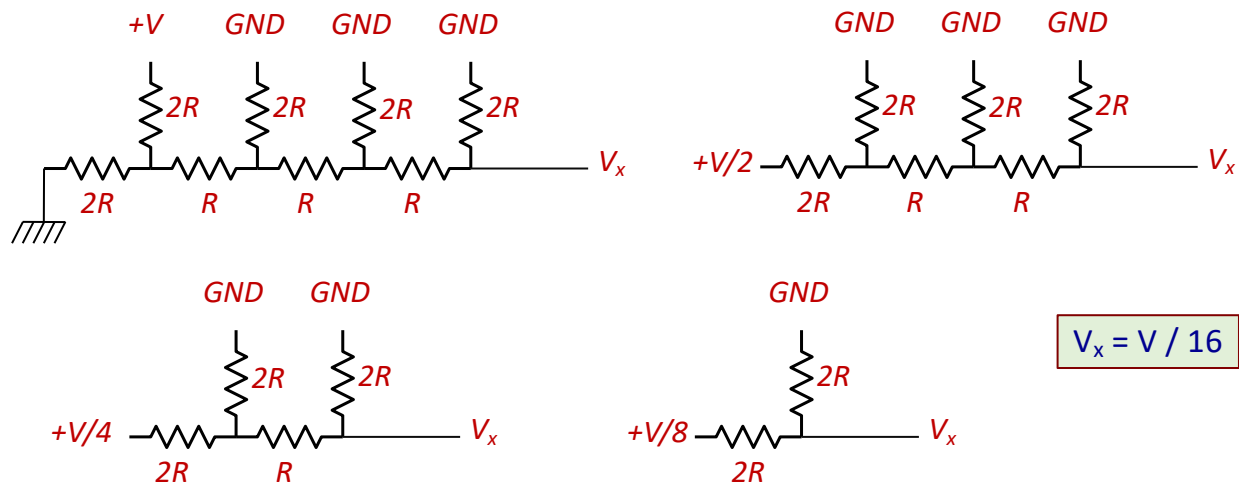
- Case 3: Input is 0010.



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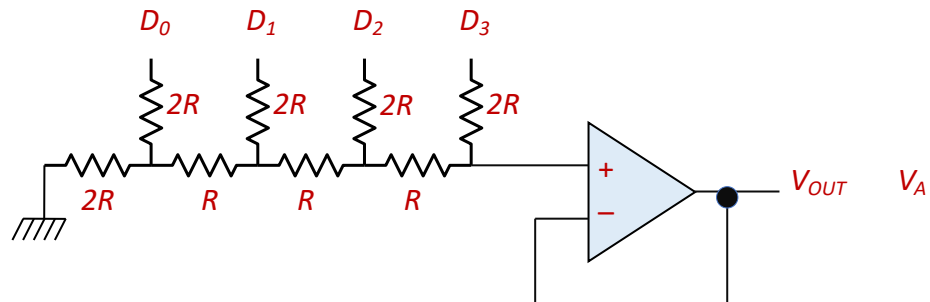
- Case 4: Input is 0001.



31

31

- When all the four inputs $D = D_3D_2D_1D_0$ are applied (where $D_i = GND$ or $+V$ volts), we can apply the principle of superposition to compute the final output voltage V_A .



$$\begin{aligned}
 V_A &= [D_3 \cdot (V/2) + D_2 \cdot (V/4) + D_1 \cdot (V/8) + D_0 \cdot (V/16)] \\
 &= [8D_3 + 4D_2 + 2D_1 + D_0] \cdot (V/16) \\
 &= D \cdot (V / 16)
 \end{aligned}$$

Thus, $V_A \propto D$.

For an N -bit DAC, the contribution of the k -th input D_k on the output voltage will be $V / 2^{N-k}$.

32

32

Analog-to-Digital Converter

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33

33

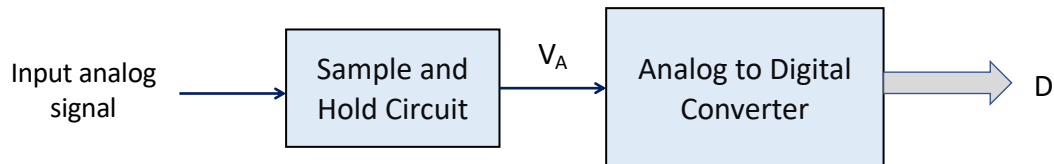
Introduction

- An analog-to-digital converter (ADC) takes an analog input V_A as input, and generates a digital word D as output, where $D \propto V_A$.
- Various types of ADC are possible:
 - a) Flash-type ADC
 - b) Counter-type ADC
 - c) Tracking-type ADC
 - d) Successive approximation ADC
 - e) Dual-slope ADC

34

34

Overall Schematic

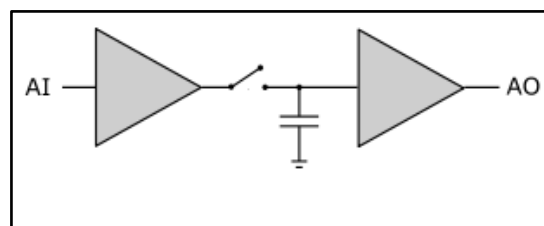


- If the input signal changes during the conversion, the final value D may be erroneous.
- We use a *sample-and-hold circuit* to sample the input voltage, and hold it to a constant value while the conversion is in progress.

35

35

Sample-and-Hold Circuit



- The input voltage AI is sampled by closing the switch; the capacitor charges to the voltage.
- During conversion, the switch is opened again, so that AO remains reasonably constant.

36

36

How Fast We Must Sample?

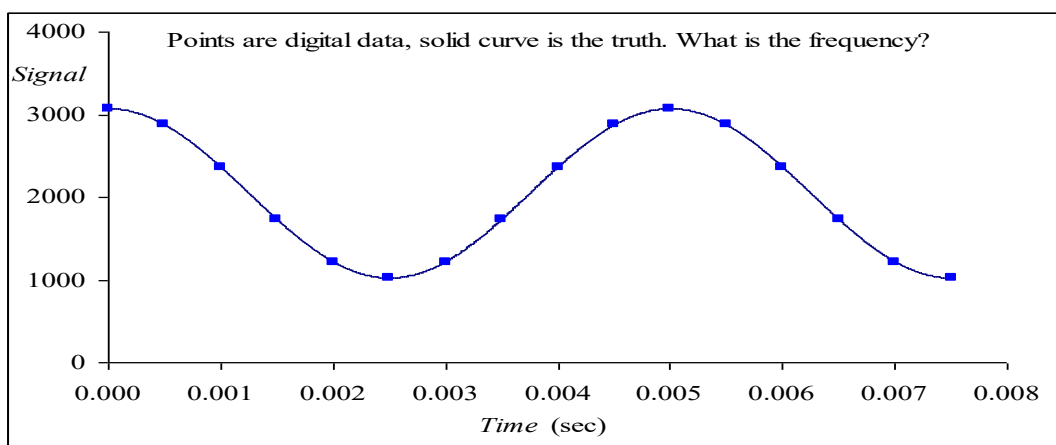
- A very important decision.
 - Guided by *Nyquist Theorem*.
- Nyquist Theorem:
 - A *band-limited* analog signal that has been sampled can be perfectly reconstructed from an infinite sequence of samples if the sampling rate f_s exceeds $2f_{max}$ samples per second, where f_{max} is the highest frequency in the original signal.
 - If the analog signal does contain frequency components larger than $f_s/2$, then there will be an *aliasing error*.
 - Aliasing is when the digital signal appears to have a different frequency than the original analog signal.

37

37

Some Sampling Examples

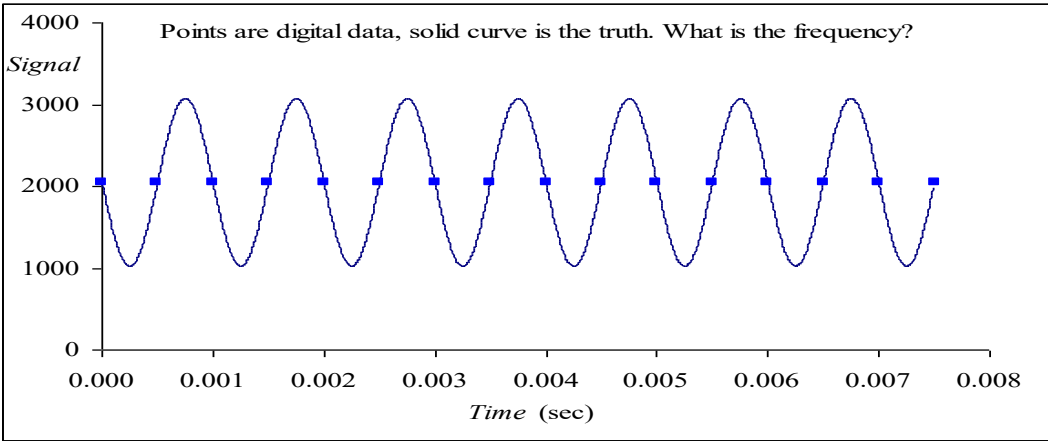
200Hz signal sampled at 2000Hz



38

38

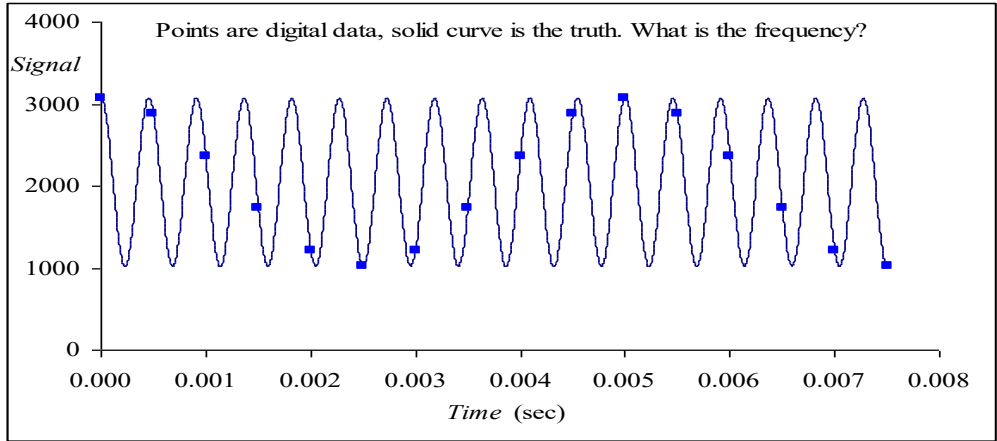
1000Hz signal sampled at 2000Hz



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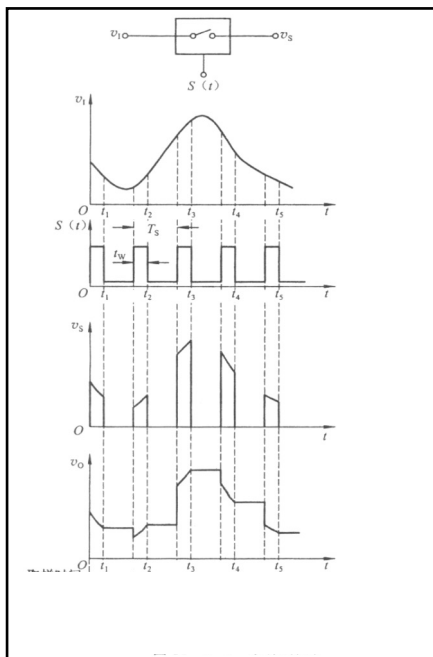
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2200Hz signal sampled at 2000Hz (*Aliasing Results*)



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40



Sampling an waveform, and reconstruction of the waveform from the sampled values.

41

41

Resolution of ADC

- The least significant bit of an analog-to-digital converter (ADC) gives the *resolution*.
- Related to full scale if the ADC is linear.
 - $LSB = A/2^n$ for an n -bit ADC.
 - For a linear 8-bit ADC with a 1V full scale input,
Resolution = $1/2^8 = 3.9 \text{ mV}$ (0.39%)

42

42

Dynamic range

- Ratio between the minimum and the maximum amplitude to be measured.
- For a linear ADC the dynamic range is related to the number of bits (i.e. the resolution).
 - An 8-bit ADC has a dynamic range of 256.
- In case of large dynamic range some non-linearity has to be introduced.

43

43

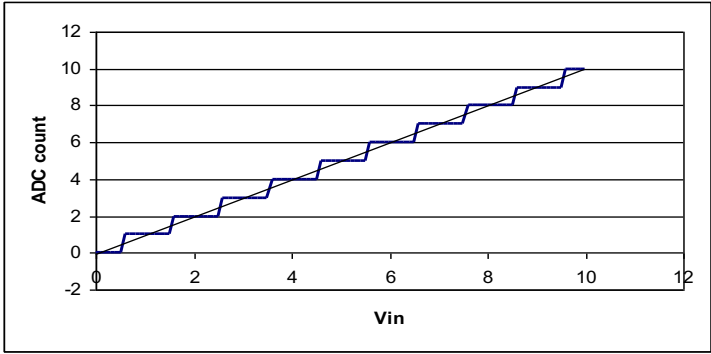
Conversion Time and Bandwidth

- How often can a conversion be done?
 - A few ns to a few ms depending on the technology.
- Input bandwidth
 - Maximum input signal bandwidth
 - Sample and hold input circuitry
 - Conversion frequency

44

44

ADC Transfer Characteristics

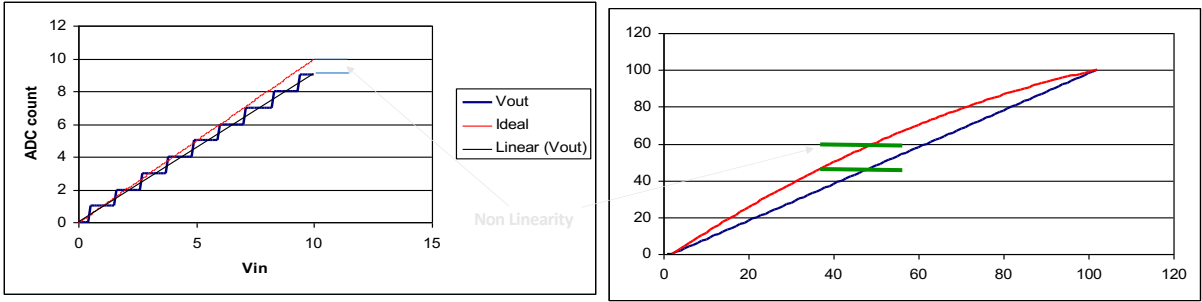


- This curve is for ideal ADC.
- Possible errors:
 - Offset
 - Integral non-linearity
 - Differential non-linearity

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Integral Non-linearity Error



- Non linearity: maximum difference between the best linear fit and the ideal curve

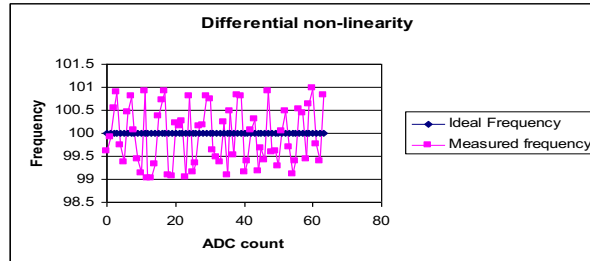
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46

Differential Non-linearity

$$1 \text{ LSB} = \frac{V_{\max}}{2^n}$$

$n\text{-bit ADC}$



- Least Significant Bit (LSB) value should be constant but it is not.
- The difference with the ideal value should not exceed 0.5 LSB.
- How to check?
 - random input covering the full range.
 - frequency histogram should be flat.

47

47

Types of ADC

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48

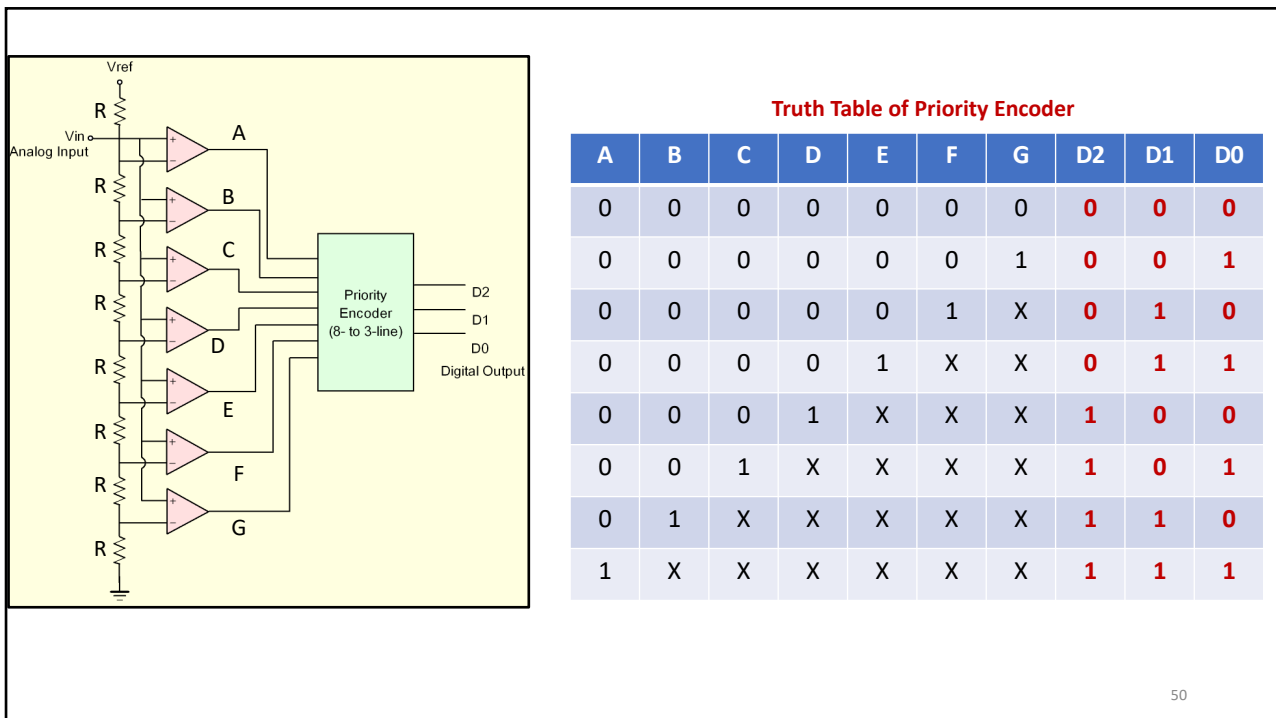
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(a) Flash-type ADC

- This type of ADC provides the fastest conversion.
- Requires large amount of hardware.
 - For an n -bit converter, we require $2^n - 1$ voltage comparators, 2^n resistors, and one 2^n -input priority encoder.
 - The output of a voltage comparator is 1 if $V^+ > V^-$, and is 0 otherwise.
- We illustrate the design for a 3-bit ADC.

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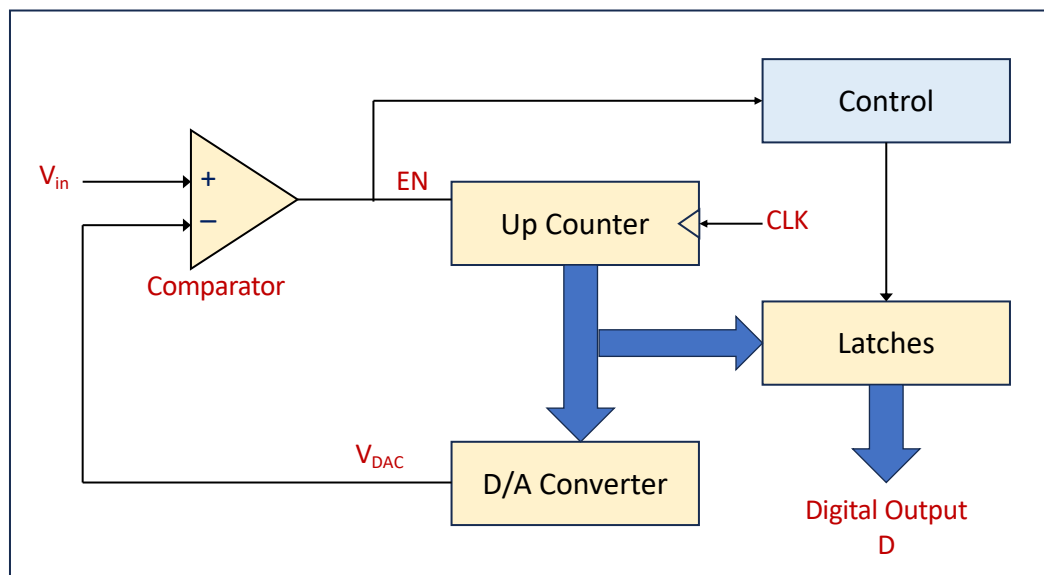
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(b) Counter-Type ADC

- A reference voltage is generated using a *binary up counter* and a *DAC*.
 - Generates a staircase waveform.
- The staircase waveform is compared with the given analog input voltage V_{in} using a comparator; as soon as the staircase voltage crosses V_{in} , the counting stops.
 - The value stored in the counter gives the digital output D .
 - The process is similar to linear search performed in hardware.

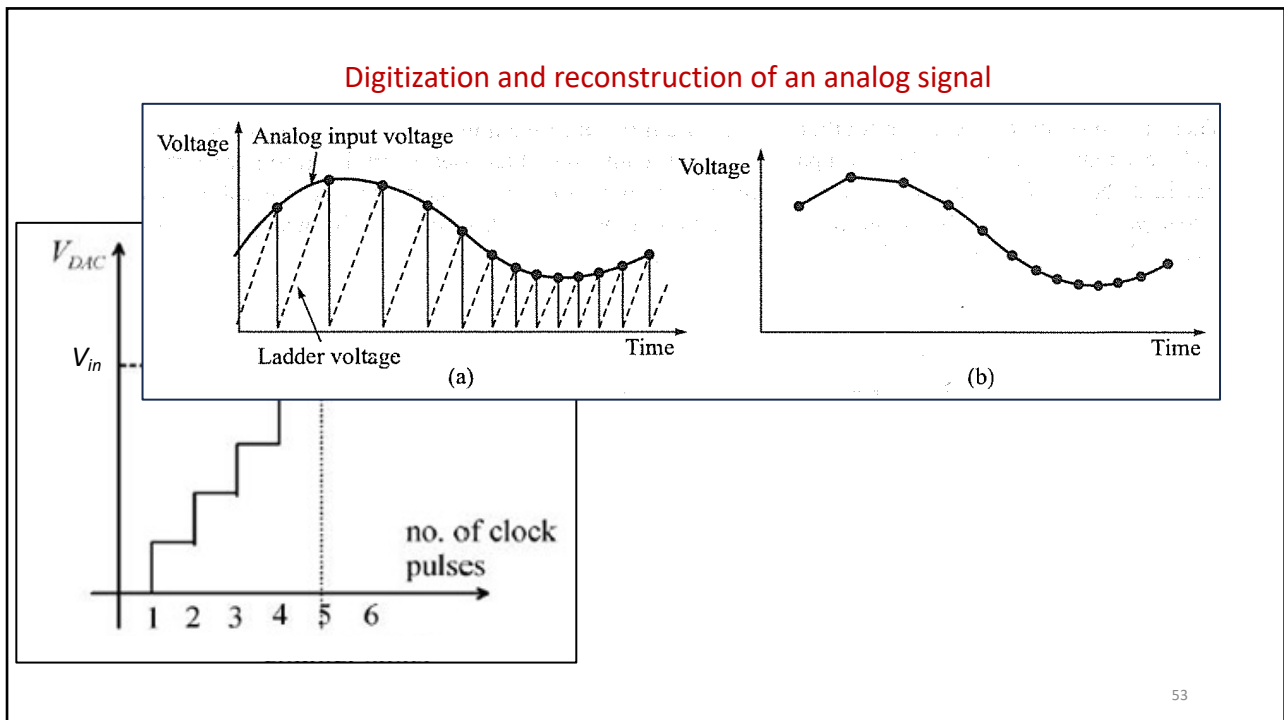
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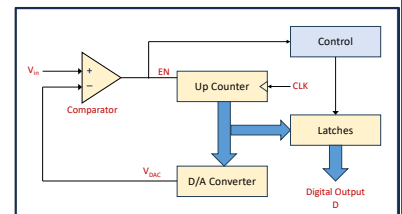
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53

Detailed Steps



- We use a D/A converter, a counter and a comparator.
 - Before a conversion starts, the counter is reset to zero.
 - The counter output is fed to the input of the DAC.
 - The DAC output D is compared with the input analog voltage V_{in} .
 - If $V_{in} > D$, then the counter is incremented by 1.
 - If $V_{in} < D$, the conversion is complete, and the counter output D gives the required digital output.
- The worst-case conversion time is equal to 2^n .

54

54

Conversion Time Calculation

- The average conversion time for an n -bit converter is $2^n / 2 = 2^{n-1}$.
- An example:
 - Consider a 10-bit converter.
 - It requires 1024 clock cycles for a full-scale count.
 - With a 1 MHz clock, the full-scale count requires 1.024 ms.
 - The average conversion time will be half of this time.

55

55

(c) Tracking-Type ADC

- Also known as *Continuous A/D Converter*.
- An obvious method for speeding up the conversion of an analog signal is to eliminate the need to reset the counter every time a conversion is made.
- This way, we would begin with the last converted point, and provide the counter with the capability to count up or down.
- The converter's digital output always tries to track the analog input; hence the name *Tracking-type ADC*.

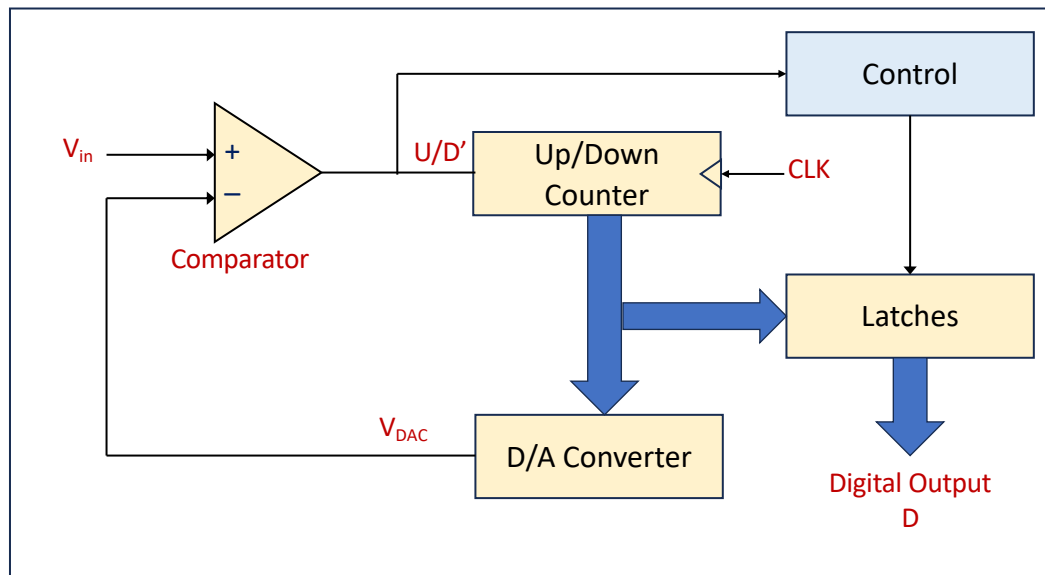
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56

- Consider an input analog signal (e.g., voice signal) that changes slowly with time.
 - Instead of starting from zero every time, it is possible to track changes in the input waveform continuously.
 - We use a *binary up/down counter*, which will be counting up or down depending on whether the input analog signal is higher or lower than the reference voltage produced by the DAC.

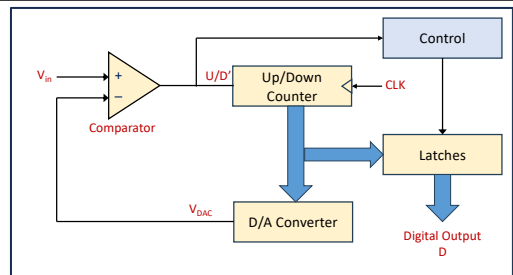
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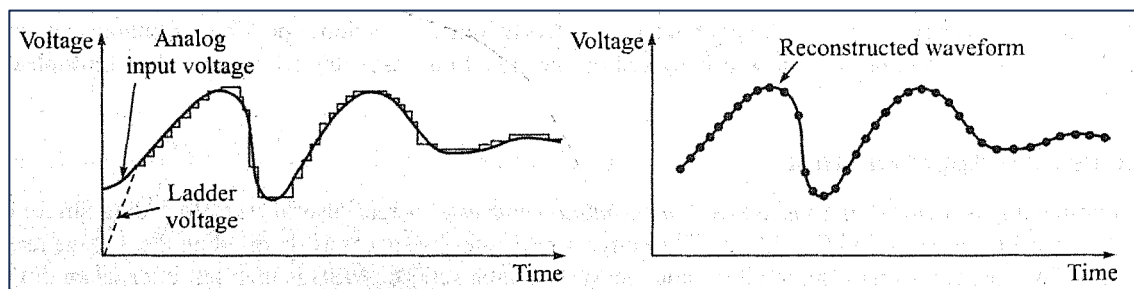
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- For the binary up-down counter:
 - If the comparator output is 1 ($V_{in} > V_{DAC}$), the counter is incremented by 1.
 - If the comparator output is 0 ($V_{in} < V_{DAC}$), the counter is decremented by 1.
 - The counter output continuously tracks the input analog voltage.
- Suitable for continuous conversion of slowly-varying waveforms.
 - If V_{in} changes too fast, the counter may not be able to track the changes.
 - Here also, the worst-case conversion time is 2^n .
 - Not suitable for fast-changing input signals.

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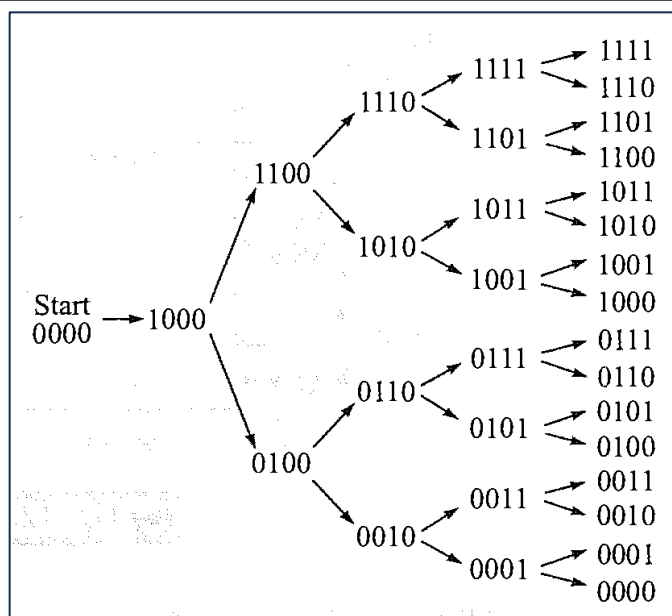
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(d) Successive Approximation Type ADC

- The principle of operation is analogous to *binary search*.
 - Suppose we are searching for a number (here V_{in}) in a sorted list.
 - We look into the middle of the list – effectively the size of the list reduces by half.
 - For 2^n steps, the number of steps required is only n .
- What modifications are required?
 - We use a *successive approximation register* (SAR) instead of a counter.
 - The SAR emulates binary search.
 - For example, in 4 bits, it starts with 1000 (i.e. 8).
 - If the input is smaller, the next value is set as 0100 (i.e. 4).
 - If the input is larger, the next value is set at 1100 (i.e. 12).

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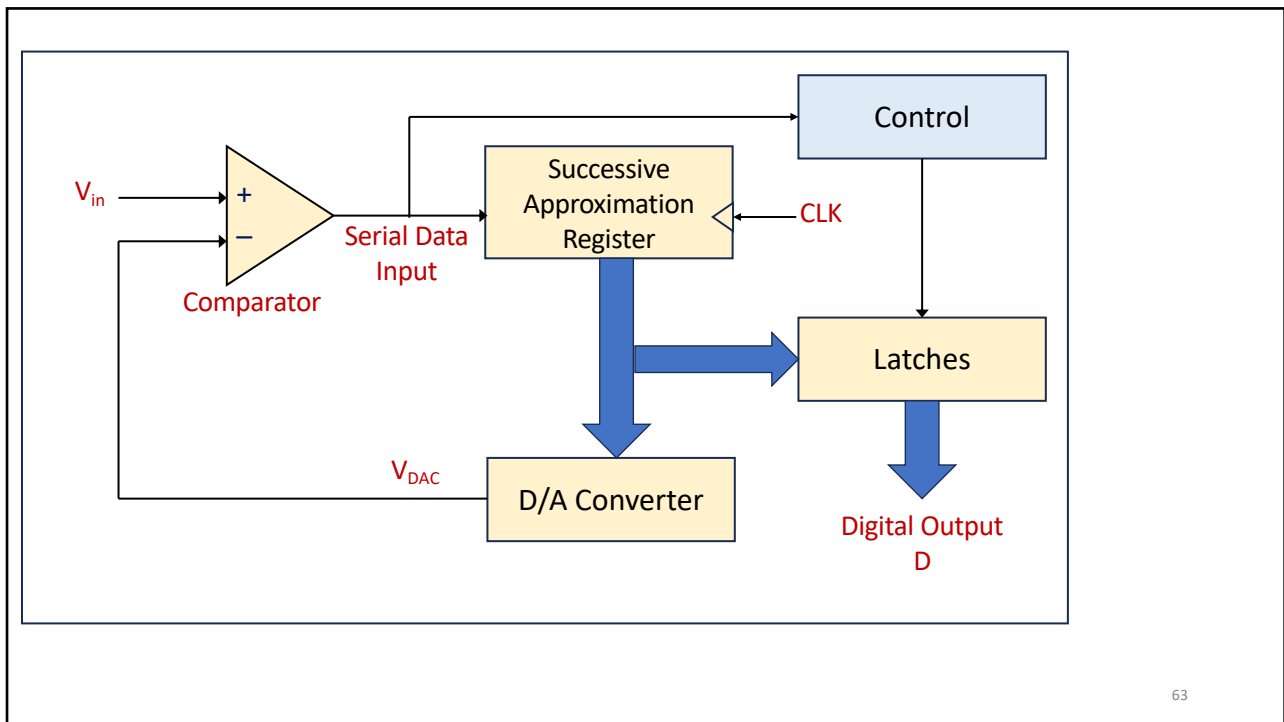
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- An example decision tree for a 4-bit ADC.
- Only 4 comparison steps are required, once per bit position.

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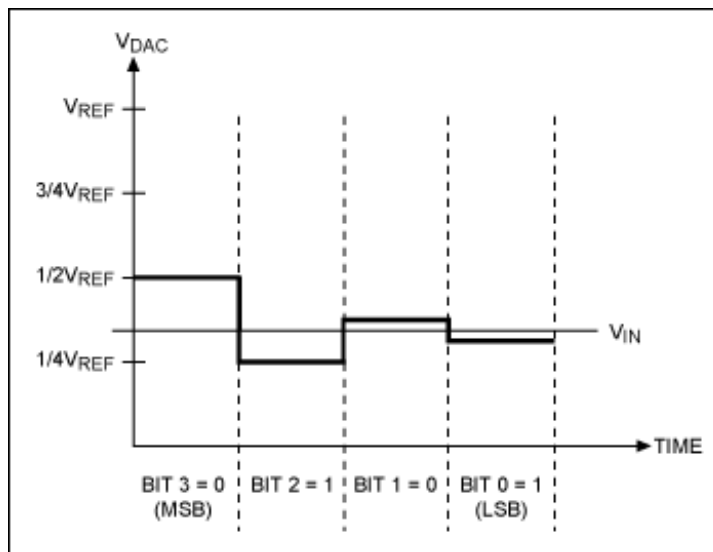
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How can the Successive Approximation Register be Designed?

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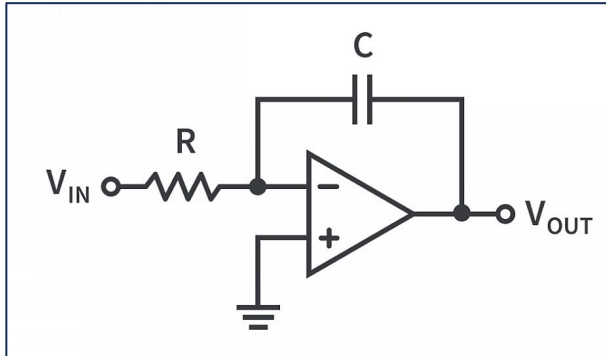
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(e) Dual-Slope Type ADC

- Here, we use a ramp generator built using an operational amplifier.
 - An integrator circuit with a constant voltage as input.
- Starting with 0 V at the integrator output, the process of conversion has two steps:
 - a) Integrate over a fixed time t_1 by applying the input analog voltage V_{in} at the integrator input.
 - b) Integrate over a variable time t_2 by applying a constant reference voltage V_R at the integrator input.

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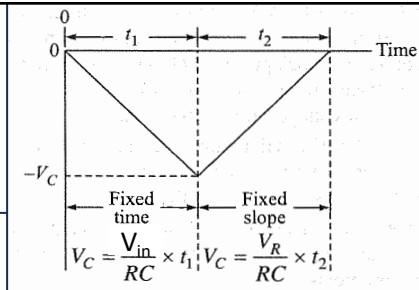


$$V_{OUT} = -\frac{1}{RC} \int_0^t V_{IN}(t) dt$$

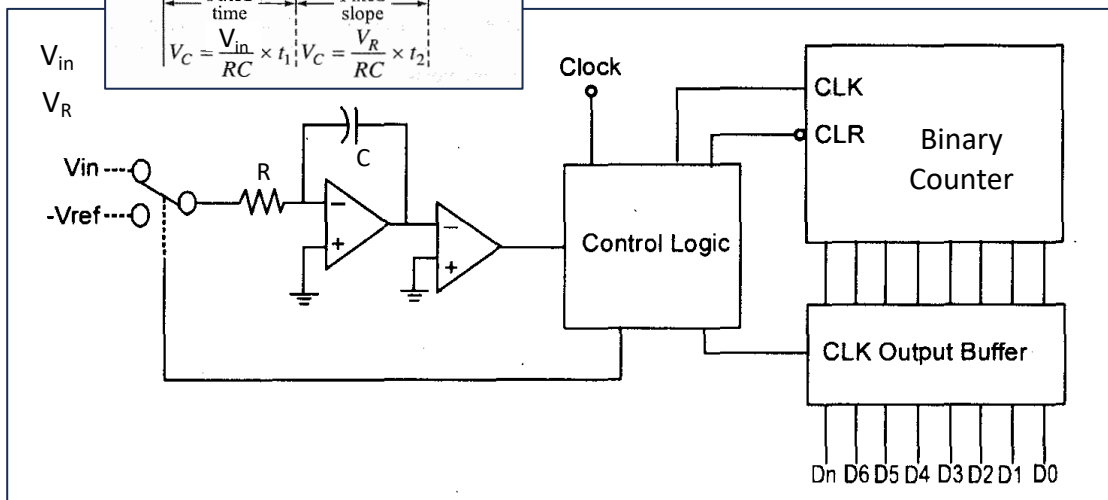
An op amp integrator

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Dual-slope ADC Schematic



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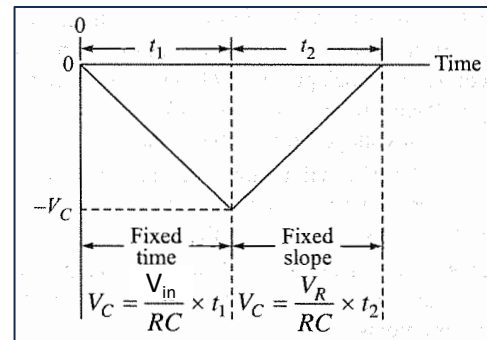
Calculation

- The integrator output voltage begins at 0 V , integrates down to $-V_C$, and then integrates back up to 0 V .

$$(V_{in} / RC) t_1 = (V_R / RC) t_2$$

$$\therefore V_{in} = V_R (t_2 / t_1)$$

- Since V_R and t_1 are constants, V_{in} will be directly proportional to t_2 .
 - The final value D stored in the counter will be proportional to t_2 , and hence to V_{in} .



69

69

Advantage of Dual-Slope ADC

- Dual slope ADCs offer exceptional accuracy and high noise immunity by averaging the input signal, making them ideal for precision measurements.
- Their key advantage is that the output is independent of component variations (resistor/capacitor drift) and clock frequency, as these errors cancel out during the dual conversion process.
 - Variations get cancelled out over the two phases of computation.
- Typically used in precision digital multimeters.

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70

70